

Практикум

A1:archive

Forty-seven questions, 135 marks. Every question the archive asks about arithmetic sequences, from May 2021 to November 2025, in the order of the nine techniques rather than the order of the papers.

О практикуме

Тема `number_algebra.sequences` взята не целиком: в ней лежат обе прогрессии сразу.

Разрез между A1 и A2 идёт по вопросу, а не по теме — у майской 2023 TZ1 бумаги пункты (a)–(c) арифметические, (d)–(e) геометрические, и это один вопрос 10. Доля 0.8 посчитана по баллам: 135 из 169.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-a1-arithmetic-sequences

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

A1 archive — arithmetic sequences and sums, all of it

Forty-seven questions, 135 marks.

Every question the archive asks about arithmetic sequences, from May 2021 to November 2025, in the order of the nine techniques rather than the order of the papers.

No theory. No worked examples. Open it, answer fifty questions, close it. The theory is in the practicum, `practicum-arithmetic-sequences.ipynb`.

§	technique	questions	marks
1	Which term is it	5	11
2	Add the first n	13	22
3	Two facts, two unknowns	3	15
4	From S_n back to u_n	4	11
5	The constant difference as a condition	6	16
6	Quantities with no index	9	25
7	The greatest sum	2	7
8	n has to be a whole number	3	10
9	Terms that are logarithms	2	18

Answers go on one line each. The checks are the same ones the practicum uses and they store nothing: a progression is handed over as a rule — a first term and a step — and the check walks it.

Three questions have no cell. *State in words, sketch a diagram and write down an expression in sigma notation* cannot be checked by anything; what the markscheme accepts for them is written out in the solutions at the end.

Two questions stop in the middle.

May 2023 TZ1 Q10 and May 2025 TZ3 Q10 begin arithmetic and finish geometric. Their later parts are in the A2 archive, and it is said again where it happens.

Solutions are at the very bottom, deliberately far away.

```

import sys
sys.path.append('.')          # from
practicum/number_algebra to practicum/kit.py
import sympy as sp           # the escape
hatch: anything not in kit is in sp
from kit import *             # checks +
progression, term, total, ln, pi, ...

language('en')                # this
notebook is in English, and so are the checks

n = symbols('n')              # the index:
every one of these papers calls it n
a, b, d, m, p, q = symbols('a b d m p q')
x = symbols('x', positive=True) # every
paper here says  $x > 1$ 

                                # k already
comes from kit

# Every progression below may be built from
an answer you have not written
# yet. The helpers return an empty
progression in that case, so the
# notebook runs top to bottom while it is
still blank.

def terms(rule):
    # the progression whose nth term is
    rule(n)
    try:
        first, second = rule(1), rule(2)
    except TypeError:
        return progression(..., ...)
    if blank(first, second):
        return progression(..., ...)

```

```

    return progression(first, second - first)

def sums(rule):
    # the progression whose first n terms add
    up to rule(n)
    try:
        first, second = rule(1), rule(2)
    except TypeError:
        return progression(..., ...)
    if blank(first, second):
        return progression(..., ...)
    return progression(first, second - 2 *
first)

def by_n(expr):
    # the progression whose nth term is your
    expression in n
    return terms(lambda i: expr.subs(n, i) if
expr is not Ellipsis else ...)

def triple(slope, constant):
    # the AS-linear definition: the slope,
    the root, the constant
    return ([Ellipsis] * 3 if blank(slope,
constant)
            else [slope, -constant / slope,
constant])

def five_of(quad):
    # the AS-quadratic definition: a, r1, b,
    r2, c
    if blank(quad):
        return [Ellipsis] * 5

```

```

    # x is declared positive above, and a
    positive x hides the negative
    # root; the roots of a quadratic are
    found on a letter that carries
    # no such promise
    z = symbols('z')
    here = quad.subs(x, z)
    lead, middle, last = Poly(here,
z).all_coeffs()
    roots = sorted(solve(here, z), key=float)
    if float(middle) < float(lead):
        roots = roots[::-1]          # the
sequence runs downhill
    return [lead, roots[0], middle, roots[1],
last]

print('ready; sympy', sp.__version__)
print('a progression:', progression(3, 2), '→
term 5 is', term(progression(3, 2), 5))

```

§ 1. Which term is it

Five questions, 11 marks. $u_n = u_1 + (n - 1)d$, read forwards and backwards.

1.1 — May 2022 TZ2 Paper 1 Q1(a), 1 mark

The n th term of an arithmetic sequence is given by $u_n = 15 - 3n$. State the value of the first term, u_1 .

1.2 — May 2022 TZ2 Paper 1 Q1(b), 2 marks

Given that the n th term of this sequence is -33 , find the value of n .

1.3 — May 2022 TZ2 Paper 1 Q1(c), 2 marks

Find the common difference, d .

```
q1_1 = ...          # the first term
q1_2 = ...          # the value of n for which
the term is -33
q1_3 = ...          # the common difference

mine = progression(q1_1, q1_3)

verify_term('1.1, 1.3', 15 - 3 * n, mine, n)
verify_term('1.2', -33, mine, q1_2)
```

1.4 — May 2021 TZ2 Paper 2 Q2(a), 2 marks

An arithmetic sequence has first term 60 and common difference -2.5 . Given that the k th term of the sequence is zero, find the value of k .

```
q1_4 = ...          # the index of the zero term

verify_term('1.4', 0, progression(60, -2.5),
q1_4)
```

1.5 — November 2025 TZ1 Paper 1 Q1(a), 4 marks

The 1st and 5th terms of an arithmetic sequence are 36 and 12 respectively. Find the 13th term of this arithmetic sequence.

```
q1_5d = ...         # the common difference
q1_5 = ...          # the 13th term

mine = progression(36, q1_5d)

verify_term('1.5 (d)', 12, mine, 5)
verify_term('1.5', q1_5, mine, 13)
```

§ 2. Add the first n

Thirteen questions, 22 marks. $S_n = \frac{n}{2}(u_1 + u_n)$, and the recurring trick of the exam: what is being added is itself an arithmetic sequence, and finding *that* is the whole of the work.

2.1 — November 2022 Paper 3 Q1(a), 1 mark

This question investigates series of the form $\sum_{i=1}^n i^q = 1^q + 2^q + 3^q + \dots + n^q$ where $n, q \in \mathbb{Z}^+$. When $q = 1$, the series is arithmetic.

Show that $\sum_{i=1}^n i = \frac{1}{2}n(n+1)$.

A "show that": the printed formula is the question's, so what is asked here is the sequence being added — its n th term.

```
q2_1 = ...      # the nth term of the series  
being added, in terms of n
```

```
verify_total('2.1', n * (n + 1) / 2,  
by_n(q2_1), n)
```

2.2 — November 2022 Paper 3 Q1(b) (i), 1 mark

Consider the case when $q = 2$. The following table gives values of n^2 and $\sum_{i=1}^n i^2$ for $n = 1, 2, 3$:

n	n^2	$\sum_{i=1}^n i^2$
1	1	1
2	4	5
3	9	p

Write down the value of p .

*The only hashed answer in the notebook:
 $1 + 4 + 9$ is not an arithmetic sequence,
and the check cannot walk it.*

```
q2_2 = ...          # the value of p  
  
check_num('2.2', q2_2, 2, '8527a891e224')
```

2.3 — November 2022 Paper 3 Q1(d) (iii), 1 mark

Using sigma notation, write down an expression for $f_q(1)$, where $f_q(x) = \sum_{i=1}^n i^q x^i$.

*No cell: the answer is a piece of notation,
and nothing here can check notation. It*

is written out in the solutions.

2.4 — May 2022 TZ1 Paper 3 Q1(a)(i), 2 marks

For an r -sided regular polygon, the n th polygonal number $P_r(n)$ is given by

$$P_r(n) = \frac{(r-2)n^2 - (r-4)n}{2}$$

For triangular numbers, verify that $P_3(n) = \frac{n(n+1)}{2}$.

Substituting $r = 3$ is the mark. What is asked here instead is the triangular number the formula then produces at $n = 10$ — the same substitution, with something to check.

```
q2_4 = ...          # the 10th triangular number

verify_total('2.4', q2_4, terms(lambda i: i),
10)
```

2.5 — May 2022 TZ1 Paper 3 Q1(a)(ii), 2 marks

The number 351 is a triangular number. Determine which one it is.

```
q2_5 = ...          # which triangular number
351 is

verify_total('2.5', 351, terms(lambda i: i),
q2_5)
```

2.6 — May 2022 TZ1 Paper 3 Q1(b)(i), 2 marks

Show that $P_3(n) + P_3(n + 1) \equiv (n + 1)^2$.

A "show that" again. What is asked here is the value both sides take at $n = 4$ — the case the next question draws.

```
q2_6 = ...          # the value of P3(4) + P3(5)

tri = terms(lambda i: i)
# 1 + 2 + 3 + ... is triangular
odds = terms(lambda i: 2 * i - 1)
# 1 + 3 + 5 + ... is square

verify_total('2.6 (as a square)', q2_6, odds,
5)
verify_total('2.6 (as two triangles)',
... if blank(q2_6) else q2_6 -
total(tri, 5), tri, 4)
```

2.7 — May 2022 TZ1 Paper 3 Q1(b)(ii), 1 mark

State, in words, what the identity given in part (b)(i) shows for two consecutive

triangular numbers.

No cell: the answer is a sentence. What the markscheme accepts is in the solutions.

2.8 — May 2022 TZ1 Paper 3 Q1(b)(iii), 1 mark

For $n = 4$, sketch a diagram clearly showing your answer to part (b)(ii).

No cell: the answer is a picture. The markscheme's own diagram is in the solutions.

2.9 — May 2022 TZ1 Paper 3 Q1(d), 3 marks

The n th pentagonal number can be represented by the arithmetic series

$$P_5(n) = 1 + 4 + 7 + \cdots + (3n - 2)$$

Hence show that $P_5(n) = \frac{n(3n - 1)}{2}$ for $n \in \mathbb{Z}^+$.

```
q2_9 = ...      # the nth term of the series  
being added, in terms of n
```

```
verify_total('2.9', n * (3 * n - 1) / 2,  
by_n(q2_9), n)
```

2.10 — May 2025 TZ1 Paper 2

Q10(a), 1 mark

Rectangular playing cards are stacked in the shape of a pyramid with n rows: a one-row pyramid is two cards, a two-row pyramid is seven. Let t_n be the number of cards used. Write down t_3 .

2.11 — May 2025 TZ1 Paper 2

Q10(b), 2 marks

Find t_4 .

2.12 — May 2025 TZ1 Paper 2 Q10(c),

3 marks

Show that $t_n = \frac{n(3n + 1)}{2}$.

Same shape as 2.9: what is asked is the number of cards row n takes on its own.

```
q2_12 = ...      # the number of cards in row
n alone, in terms of n
q2_10 = ...      # t3
q2_11 = ...      # t4

rows = by_n(q2_12)

verify_total('2.12', n * (3 * n + 1) / 2,
rows, n)
verify_total('2.10', q2_10, rows, 3)
verify_total('2.11', q2_11, rows, 4)
```

2.13 — November 2025 TZ1 Paper 1

Q1(b), 2 marks

The sum of the first n terms of the arithmetic sequence of question 1.5 is zero. Find the value of n .

```
q2_13 = ...      # the number of terms whose  
sum is zero
```

```
verify_total('2.13', 0, progression(36, -6),  
q2_13)
```

§ 3. Two facts, two unknowns

Three questions, 15 marks. Neither u_1 nor d is given, and the paper hands you exactly two facts about them. Every check below is backwards: the answers go inside the progression, and the question's own numbers come back out.

3.1 — *May 2021 TZ1 Paper 1 Q2, 5 marks*

Consider an arithmetic sequence where $u_8 = S_8 = 8$. Find the value of the first term, u_1 , and the value of the common difference, d .

```
q3_lu = ...      # the first term
q3_ld = ...      # the common difference

mine = progression(q3_lu, q3_ld)

verify_term('3.1 (u8)', 8, mine, 8)
verify_total('3.1 (S8)', 8, mine, 8)
```

3.2 — November 2025 TZ3 Paper 1

Q1, 5 marks

The 7th term of an arithmetic sequence is 6. The sum of the 6th term and the 12th term is 24. Find the first term and the common difference.

```
q3_2u = ...      # the first term
q3_2d = ...      # the common difference

mine = progression(q3_2u, q3_2d)
rest = ... if blank(q3_2u, q3_2d) else 24 -
term(mine, 12)

verify_term('3.2 (u7)', 6, mine, 7)
verify_term('3.2 (u6 + u12)', rest, mine, 6)
```

3.3 — November 2023 TZ1 Paper 1

Q3(a), 5 marks

The sum of the first n terms of an arithmetic sequence is given by $S_n = pn^2 - qn$, where p and q are positive constants. It is given that $S_4 = 40$ and $S_5 = 65$. Find the value of p and the value of q .

The same paper was set in two zones that November, word for word. It is counted once.

```
q3_3p = ...      # p
q3_3q = ...      # q

mine = sums(lambda i: q3_3p * i ** 2 - q3_3q
* i)

verify_total('3.3 (S4)', 40, mine, 4)
verify_total('3.3 (S5)', 65, mine, 5)
```

§ 4. From S_n back to u_n

Four questions, 11 marks. $u_1 = S_1$ and $u_n = S_n - S_{n-1}$.

4.1 — May 2023 TZ1 Paper 1 Q10(a), 4 marks

Consider the arithmetic sequence $u_1, u_2, u_3, \dots$. The sum of the first n terms of this sequence is given by $S_n = n^2 + 4n$.

- (i)** Find the sum of the first five terms.
- (ii)** Given that $S_6 = 60$, find u_6 .

4.2 — May 2023 TZ1 Paper 1 Q10(b), 2 marks

Find u_1 .

4.3 — May 2023 TZ1 Paper 1 Q10(c), 3 marks

Hence or otherwise, write an expression for u_n in terms of n .

Parts (d) and (e) of this question turn to a geometric sequence and are in the A2 archive.

```

q4_1s = ...      # S5
q4_1u = ...      # u6
q4_2 = ...       # u1
q4_3 = ...       # u_n, in terms of n

given = sums(lambda i: i ** 2 + 4 * i)

verify_total('4.1(i)', q4_1s, given, 5)
verify_term('4.1(ii)', q4_1u, given, 6)
verify_term('4.2', q4_2, given, 1)
verify_term('4.3', q4_3, given, n)

```

4.4 — November 2023 TZ1 Paper 1 Q3(b), 2 marks

For the sequence of question 3.3, find the value of u_5 .

```

q4_4 = ...      # u5

verify_term('4.4', q4_4, sums(lambda i: 3 * i
** 2 - 2 * i), 5)

```

§ 5. The constant difference as a condition

Six questions, 16 marks. $u_2 - u_1 = u_3 - u_2$, imposed to find a letter or verified to prove a family arithmetic.

5.1 — May 2025 TZ3 Paper 1 Q10(a) (i), 3 marks

Consider the sequence $\{u_n\}$ whose first three terms are $u_1 = k - 5$, $u_2 = 3 - 2k$ and $u_3 = 5k + 3$, where $k \in \mathbb{R}$. Consider the case when $\{u_n\}$ is arithmetic. Find the value of k .

5.2 — May 2025 TZ3 Paper 1 Q10(a) (ii), 2 marks

Hence, or otherwise, find u_3 .

Parts (b) and (c) of this question take the same three terms and make them geometric. They are in the A2 archive.

```

q5_1 = ...          # the value of k
q5_2 = ...          # the third term

three = ([Ellipsis] * 3 if blank(q5_1)
         else [q5_1 - 5, 3 - 2 * q5_1, 5 *
               q5_1 + 3])

verify_arithmetic('5.1', three)
verify_term('5.2', q5_2, terms(lambda i:
three[i - 1]), 3)

```

5.3 — May 2024 TZ2 Paper 1 Q10(a), 2 marks

Consider the arithmetic sequence $a, p, q, \dots$, where $a, p, q \neq 0$. Show that $2p - q = a$.

A "show that": what is asked here is the same identity solved for the middle term.

```

q5_3 = ...          # p in terms of a and q

verify_arithmetic('5.3', [a, q5_3, q])

```

5.4 — May 2024 TZ2 Paper 1 Q10(c), 2 marks

The first term of both that arithmetic sequence and a geometric sequence $a, s, t, \dots$ is a . It is given that $q = t = 1$. Show that $p > \frac{1}{2}$.

The reasoning runs through a : with $q = 1$ the identity of 5.3 gives a in terms of p , and $a = s^2 > 0$ finishes it. What is asked here is that expression for a .

```
q5_4 = ...          # a in terms of p, when q =
1

verify_arithmetic('5.4', [q5_4, p, 1])
```

5.5 — May 2024 TZ2 Paper 1 Q10(d), 4 marks

Consider the case where $a = 9$, $s > 0$ and $q = t = 1$. Write down the first four terms of **(i)** the arithmetic sequence; **(ii)** the geometric sequence.

Only part (i) is here. The geometric four are in the A2 archive.

```
q5_5 = [...]        # the first four terms of
the arithmetic sequence

verify_start('5.5', q5_5, terms(lambda i: [9,
5, 1][i - 1]
                                if not
blank(q5_5) else ...))
verify_arithmetic('5.5 (arithmetic)', q5_5)
```

5.6 — May 2023 TZ2 Paper 1 Q12(d), 3 marks

The regions bounded by the curve $y = \cos \sqrt{x}$ and the x -axis are $R_1, R_2, R_3, \dots$, and part (c) of the paper gives their areas as $R_n = 4n\pi$. Hence, show that the areas form an arithmetic sequence.

What is asked here is the common difference, and the check refuses to look at it before the difference is constant.

```
q5_6 = ...          # the common difference of
the areas

verify_step('5.6', q5_6, [4 * i * pi for i in
(1, 2, 3, 4)])
```

§ 6. Quantities with no index

Nine questions, 25 marks — one Paper 3 investigation from end to end. May 2023 TZ2 Paper 3 Q2 puts a coefficient, a root and a constant into arithmetic sequence and builds thirty-one marks on that one condition.

Throughout: $L(x) = mx + c$ with root r is **AS-linear** when m, r, c in that order are in arithmetic sequence; $Q(x) = ax^2 + bx + c$ with roots r_1, r_2 is **AS-quadratic** when a, r_1, b, r_2, c in that order are.

6.1 — May 2023 TZ2 Paper 3 Q2(a), 2 marks

Show that $L(x) = 2x - 1$ is an AS-linear function.

What is asked is the common difference of the sequence it makes.

```
q6_1 = ...          # the common difference of
the sequence made by 2x - 1

verify_step('6.1', q6_1, triple(2, -1))
```

6.2 — May 2023 TZ2 Paper 3 Q2(b) (ii), 4 marks

Given that $L(x) = mx + c$ is an AS-linear function, show that $L(x) = mx - \frac{m^2}{m+2}$.

What is asked is c in terms of m . Your expression goes into the triple $(m, -c/m, c)$, and the triple has to be arithmetic at every m the check tries.

```
q6_2 = ...          # c in terms of m

verify_arithmetic('6.2', triple(m, q6_2), m,
(1, 3, -5))
```

6.3 — May 2023 TZ2 Paper 3 Q2(b) (iii), 1 mark

State any further restrictions on the value of m (beyond $m, c \neq 0$).

Hashed: the answer is the one value of m at which your expression from 6.2 has no value, and there is no sequence left to walk.

```
q6_3 = ...          # the excluded value of m

check_num('6.3', q6_3, 2, 'cf3bae39dd69')
```

6.4 — May 2023 TZ2 Paper 3 Q2(e)(i), 1 mark

Let $Q(x) = ax^2 + bx + c$ have roots r_1, r_2 , and suppose Q is AS-quadratic. Write down an expression for $r_2 - r_1$ in terms of a and b .

```
q6_4 = ...          # r2 - r1, in terms of a and b

# your gap, spread over the five quantities
a, r1, b, r2, c
chain = ([Ellipsis] * 5 if blank(q6_4)
         else [a, a + q6_4 / 2, b, b + q6_4 / 2, b + q6_4])

verify_arithmetic('6.4', chain)
```

6.5 — May 2023 TZ2 Paper 3 Q2(e)(ii), 2 marks

Use your answers to parts (d)(i) and (e)(i) to show that $r_1 = \frac{a^2 - ab - b}{2a}$.

Part (d)(i) of the paper is Vieta's $r_1 + r_2 = -\frac{b}{a}$. What is asked here is r_2 , which the printed r_1 and that relation pin down together.

```

q6_5 = ...          # r2, in terms of a and b

# Vieta, from part (d)(i) of the same paper:
r1 + r2 = -b/a
pair = ... if blank(q6_5) else q6_5 + (a ** 2
- a * b - b) / (2 * a)

verify_exact('6.5', pair, -b / a)

```

6.6 — May 2023 TZ2 Paper 3 Q2(e) (iii), 3 marks

Use the result from part (e)(ii) to show that $b = 0$ or $a = -\frac{1}{2}$.

What is asked is the second of those: the value of a at which a, r_1, b is arithmetic whatever b is.

```

q6_6 = ...          # the value of a that works
for every b

root_one = ... if blank(q6_6) else (q6_6 ** 2
- q6_6 * b - b) / (2 * q6_6)

verify_arithmetic('6.6', [q6_6, root_one, b],
b, (1, 2, 5))

```

6.7 — May 2023 TZ2 Paper 3 Q2(f), 5 marks

Consider the case where $b = 0$. Determine the two AS-quadratic functions that

satisfy this condition.

```
q6_7 = [...]      # the two quadratics, as
expressions in x

for got in q6_7:
    verify_arithmetic('6.7', five_of(got))
```

6.8 — May 2023 TZ2 Paper 3 Q2(g)(i), 2 marks

Now consider the case where $a = -\frac{1}{2}$. Find an expression for r_1 in terms of b .

```
q6_8 = ...        # r1 in terms of b

verify_arithmetic('6.8', [Rational(-1, 2),
q6_8, b], b, (1, 3, -5))
```

6.9 — May 2023 TZ2 Paper 3 Q2(g)(ii), 5 marks

Hence or otherwise, determine the exact values of b and c such that AS-quadratic functions are formed. Give your answers in the form $\frac{-p \pm q\sqrt{s}}{2}$ where $p, q, s \in \mathbb{Z}^+$.

```
q6_9b = [...]    # the two values of b
q6_9c = [...]    # the two values of c, in
the same order

for got_b, got_c in zip(q6_9b, q6_9c):
    quad = (... if blank(got_b, got_c)
            else -x ** 2 / 2 + got_b * x +
got_c)
    verify_arithmetic('6.9', five_of(quad))
    verify_exact('6.9 (exact)', got_b, got_b)
# a decimal earns nothing here
```

§ 7. The greatest sum

Two questions, 7 marks. The sum grows exactly while the terms are positive.

7.1 — May 2021 TZ2 Paper 2 Q2(b), 3 marks

For the sequence of question 1.4 — first term 60, common difference -2.5 — let S_n denote the sum of the first n terms. Find the maximum value of S_n .

```
q7_1 = ...           # the maximum value of Sn

verify_peak('7.1', q7_1, progression(60,
-2.5))
```

7.2 — May 2025 TZ2 Paper 2 Q7(b), 4 marks

A geometric sequence has first term 80 and fourth term 0.74088; its second term is 16.8. The first two terms of this geometric sequence are also the first term and eleventh term respectively of an arithmetic sequence. Let S_n denote the sum of the first n terms of the

arithmetic sequence. Find the greatest value of S_n , giving your answer to two decimal places.

Part (a) — finding the second term of the geometric sequence — is in the A2 archive.

```
q7_2d = ...      # the common difference of
the arithmetic sequence
q7_2 = ...      # the greatest value of  $S_n$ 

mine = progression(80, q7_2d)

verify_term('7.2 (d)', 16.8, mine, 11)
verify_peak('7.2', q7_2, mine)
```

§ 8. n has to be a whole number

Three questions, 10 marks. The only place in the topic where algebra does not work and every markscheme says *uses a table of values*.

8.1 — May 2022 TZ1 Paper 3 Q1(e), 5 marks

By using a suitable table of values or otherwise, determine the smallest positive integer, greater than 1, that is both a triangular number and a pentagonal number.

Which triangular number and which pentagonal number it is are asked for too: without them the check has nothing to test but the answer itself.

```

q8_1 = ...          # the number
q8_li = ...         # which triangular number it
is
q8_lj = ...         # which pentagonal number it
is

verify_total('8.1 (triangular)', q8_1,
terms(lambda i: i), q8_li)
verify_total('8.1 (pentagonal)', q8_1,
terms(lambda i: 3 * i - 2), q8_lj)

```

8.2 — May 2025 TZ1 Paper 2 Q10(e), 2 marks

A complete pyramid stack of playing cards is created using cards taken from full packs of 52 with no cards left over. Find the minimum number of rows in this stack.

*How many packs it uses is asked for the same reason. That yours is the **smallest** such stack the check cannot confirm — the table is what earns that mark.*

```

q8_2 = ...          # the minimum number of rows
q8_2k = ...         # how many full packs of 52
that stack uses

cards = ... if blank(q8_2k) else 52 * q8_2k

verify_total('8.2', cards, terms(lambda i: 3
* i - 1), q8_2)

```

8.3 — May 2023 TZ2 Paper 3 Q2(c), 3 marks

There are only three **integer** sets of values of m , r and c that form an AS-linear function. One of these is $L(x) = -x - 1$. Use part (b) to determine the other two AS-linear functions with integer values of m , r and c .

```

q8_3 = [...]      # the other two AS-linear
functions, as expressions in x

for got in q8_3:
    made = (triple(got.coeff(x, 1),
got.coeff(x, 0))
            if got is not Ellipsis else
[...])
    # rounding leaves a whole number alone
and moves any other, so the
    # check reports which of m, r, c is not
an integer
    rounded = ([...] if blank(made[0])
                else [floor(v + Rational(1,
2)) for v in made])

    verify_arithmetic('8.3', made)
    verify_start('8.3 (whole numbers)',
rounded,
                progression(made[0], made[1]
- made[0])
                if not blank(made[0]) else
progression(..., ...))

```

§ 9. Terms that are logarithms

Two questions, 18 marks — and both are Paper 1, where a decimal brings nothing at all.

9.1 — May 2022 TZ1 Paper 1 Q10(b), 12 marks

Consider the series $\ln x + p \ln x + \frac{1}{3} \ln x + \dots$, where $x \in \mathbb{R}$, $x > 1$ and $p \in \mathbb{R}$, $p \neq 0$. Now consider the case where the series is arithmetic with common difference d .

(i) Show that $p = \frac{2}{3}$. **(ii)** Write down d in the form $k \ln x$, where $k \in \mathbb{Q}$. **(iii)** The sum of the first n terms of the series is $\ln\left(\frac{1}{x^3}\right)$. Find the value of n .

Part (a) of the paper takes the same series and makes it geometric. It is in the A2 archive.

```

q9_lp = ...      # p
q9_ld = ...      # the common difference, in
the form k*ln(x)
q9_ln = ...      # the number of terms

three = [...] if blank(q9_lp) else [ln(x),
q9_lp * ln(x), ln(x) / 3]

verify_arithmetic('9.1(i)', three)
verify_step('9.1(ii)', q9_ld, three)
verify_total('9.1(iii)', ln(1 / x ** 3),
progression(ln(x), q9_ld), q9_ln)

```

9.2 — May 2024 TZ2 Paper 1 Q10(e), 6 marks

The arithmetic and the geometric sequences of question 5.5 are used to form a new arithmetic sequence u_n , whose first three terms are

$$u_1 = 9 + \ln 9,$$

$$u_2 = 5 + \ln 3,$$

$$u_3 = 1 + \ln 1$$

(i) Find the common difference of the new sequence in terms of $\ln 3$. **(ii)** Show

that $\sum_{i=1}^{10} u_i = -90 - 25 \ln 3$.

```
q9_2d = ...      # the common difference
q9_2s = ...      # the sum of the first ten
terms

verify_step('9.2(i)', q9_2d, [9 + ln(9), 5 +
ln(3), 1 + ln(1)])
verify_total('9.2(ii)', q9_2s, progression(9
+ ln(9), q9_2d), 10)
verify_exact('9.2 (exact)', q9_2s, -90 - 25 *
ln(3))
```



Solutions

§ 1. Which term is it

1.1 $u_1 = 15 - 3(1) = \boxed{12}$. Not 15: that is u_0 .

1.2 $15 - 3n = -33 \Rightarrow 3n = 48 \Rightarrow \boxed{n = 16}$.

1.3 $u_2 - u_1 = 9 - 12 = \boxed{-3}$; the coefficient of n is the common difference, and the markscheme accepts "*recognize gradient is -3* " alone.

1.4 $60 - 2.5(k - 1) = 0 \Rightarrow k - 1 = 24 \Rightarrow \boxed{k = 25}$.

1.5 Four steps from u_1 to u_5 : $12 = 36 + 4d \Rightarrow d = -6$. Then $u_{13} = 36 + 12(-6) = \boxed{-36}$, or $u_5 + 8(-6) = 12 - 48 = -36$.

§ 2. Add the first n

2.1 The series is $1 + 2 + 3 + \dots$, so its n th term is $\boxed{n}$, and $S_n = \frac{n}{2}(1 + n) = \frac{1}{2}n(n + 1)$.

2.2 $p = 1 + 4 + 9 = \boxed{14}$.

2.3 $f_q(1) = 1^q + 2^q + \dots + n^q$, so

$$f_q(1) = \boxed{\sum_{i=1}^n i^q}$$

The markscheme also accepts $\sum_{i=1}^n 1^i i^q$.

2.4 Substituting $r = 3$:

$$\begin{aligned} P_3(n) &= \frac{(3-2)n^2 - (3-4)n}{2} \\ &= \frac{n^2 + n}{2} \\ &= \frac{n(n+1)}{2} \end{aligned}$$

and at $n = 10$ that is $\frac{10 \cdot 11}{2} = \boxed{55}$.

The markscheme is explicit that numerical verification earns nothing — the substitution has to be shown.

2.5 $\frac{n(n+1)}{2} = 351 \Rightarrow n^2 + n - 702 = 0 \Rightarrow (n - 26)(n + 27) = 0$, so $\boxed{n = 26}$. Giving " $n = -27, 26$ " is marked wrong: an index is positive.

2.6 $P_3(4) + P_3(5) = 10 + 15 = \boxed{25} = 5^2$, which is $(n+1)^2$ at $n = 4$. In general

$$\begin{aligned}
& P_3(n) \\
& \quad + P_3(n+1) \qquad = \frac{n(n+1)}{2} \\
& \quad + \frac{(n+1)(n+2)}{2} \\
& = \frac{(n+1)(2n+2)}{2} \\
& = (n+1)^2
\end{aligned}$$

2.7 *The sum of the n th and $(n+1)$ th triangular numbers is the $(n+1)$ th square number.* That sentence is the whole mark.

2.8 The markscheme's diagram is a 5×5 square split by its diagonal:

```

X X X X X
O X X X X
O O X X X
O O O X X
O O O O X

```

fifteen X's (the 5th triangular number), ten O's (the 4th), twenty-five in all (the 5th square number). Any equivalent picture showing $P_3(4)$, $P_3(5)$ and $P_4(5)$ is accepted.

2.9 The series $1 + 4 + 7 + \dots$ has $u_1 = 1$ and $d = 3$, so its n th term is $\boxed{3n - 2}$ and

$$P_5(n) = \frac{n}{2}(1 + (3n - 2)) = \frac{n(3n - 1)}{2}$$

2.10 $t_3 = 2 + 5 + 8 = \boxed{15}$.

2.11 $t_4 = 15 + 11 = \boxed{26}$. The fourth row is eleven cards, not ten.

2.12 Row n takes $\boxed{3n - 1}$ cards — two more slanted and one more horizontal than the row above — so $t_n = \frac{n}{2}(2 + (3n - 1)) = \frac{n}{2}(3n + 1)$.

2.13 $\frac{n}{2}(36 + u_n) = 0$ with $n \neq 0$ needs $u_n = -36$, which by 1.5 is the 13th term: $\boxed{n = 13}$. By symmetry: $36 + 30 + \cdots + 0 + \cdots - 30 - 36$ has $6 + 1 + 6$ terms.

The markscheme condones $n = 0, 13$. Zero terms do add to zero — but an index counts terms.

§ 3. Two facts, two unknowns

3.1 $u_1 + 7d = 8$ and $\frac{8}{2}(2u_1 + 7d) = 8$, that is $2u_1 + 7d = 2$. Subtracting, $u_1 = -6$; then $7d = 14$ and $d = 2$.

$$\boxed{u_1 = -6, \quad d = 2}$$

Shorter: $S_8 = 4(u_1 + u_8) = 4(u_1 + 8) = 8$ gives $u_1 = -6$ at once, because u_8 is already known.

3.2 $u_1 + 6d = 6$, and $u_6 + u_{12} = (6 - d) + (6 + 5d) = 12 + 4d = 24$, so $d = 3$ and $u_1 = 6 - 18 = -12$.

$$\boxed{u_1 = -12, \quad d = 3}$$

Measuring u_6 and u_{12} from the known u_7 rather than from u_1 is the markscheme's second method and three lines shorter.

3.3 $16p - 4q = 40$ and $25p - 5q = 65$, that is $4p - q = 10$ and $5p - q = 13$. Subtracting, $\boxed{p = 3}$, then $\boxed{q = 2}$.

Or through u_1 and d : $S_4 = 2(2u_1 + 3d) = 40$ and $S_5 = \frac{5}{2}(2u_1 + 4d) = 65$ give $u_1 = 1$, $d = 6$, and $S_n = \frac{n}{2}(2 + 6(n - 1)) = 3n^2 - 2n$.

§ 4. From S_n back to u_n

4.1(i) $S_5 = 25 + 20 = \boxed{45}$.

(ii) $u_6 = S_6 - S_5 = 60 - 45 = \boxed{15}$.

4.2 $u_1 = S_1 = 1 + 4 = \boxed{5}$.

$$\mathbf{4.3} \quad u_n = S_n - S_{n-1} = (n^2 + 4n) - ((n-1)^2 + 4(n-1)) = \boxed{2n+3}.$$

Or through d : $u_2 = S_2 - S_1 = 12 - 5 = 7$, so $d = 7 - 5 = 2$ and $u_n = 5 + 2(n-1)$. Same answer, and the check accepts either — it walks the sequence rather than reading the expression.

$$\mathbf{4.4} \quad u_5 = S_5 - S_4 = 65 - 40 = \boxed{25}. \text{ Nothing about } p \text{ and } q \text{ is needed; this is the fastest mark on that paper.}$$

§ 5. The constant difference as a condition

$$\mathbf{5.1} \quad (3 - 2k) - (k - 5) = (5k + 3) - (3 - 2k) \text{ gives } 8 - 3k = 7k, \text{ so } \boxed{k = \frac{4}{5}}. \text{ The mean form is quicker: } \frac{(k-5)+(5k+3)}{2} = 3 - 2k \Rightarrow 3k - 1 = 3 - 2k.$$

$$\mathbf{5.2} \quad u_3 = 5\left(\frac{4}{5}\right) + 3 = \boxed{7}. \text{ The three terms are } -\frac{21}{5}, \frac{7}{5}, 7, \text{ with difference } \frac{28}{5}.$$

$$\mathbf{5.3} \quad p - a = q - p, \text{ so } \boxed{p = \frac{a+q}{2}} - \text{ which rearranges to } 2p - q = a.$$

5.4 With $q = 1$: $a = 2p - 1$, so $\boxed{a = 2p - 1}$. Since $a = s^2$ and $s > 0$, $a > 0$, hence $2p - 1 > 0$ and $p > \frac{1}{2}$.

5.5 $a = 9$, $q = 1$, so $p = \frac{9+1}{2} = 5$ and $d = -4$:

$$\boxed{9, 5, 1, -3}$$

(The geometric four are $9, 3, 1, \frac{1}{3}$ — A2.)

5.6 $R_{n+1} - R_n = 4(n+1)\pi - 4n\pi = \boxed{4\pi}$, **which does not depend on n , so the difference is constant.** That last clause is the R1, and the markscheme awards M0 for checking R_3 and R_2 instead.

§ 6. Quantities with no index

6.1 $L(x) = 2x - 1$ has $m = 2$, $c = -1$, $r = \frac{1}{2}$. The sequence $2, \frac{1}{2}, -1$ has $d = \boxed{-\frac{3}{2}}$ both ways, so it is arithmetic.

6.2 With $r = -\frac{c}{m}$, the condition $r - m = c - r$ gives $-\frac{2c}{m} = c + m$, so $m^2 + cm + 2c = 0$ and

$$c = \boxed{-\frac{m^2}{m+2}}$$

6.3 c has no value at $m+2=0$, so $\boxed{m \neq -2}$.

6.4 In a, r_1, b, r_2, c the gap $r_2 - r_1$ spans two steps, and so does $b - a$:

$$r_2 - r_1 = \boxed{b - a}$$

6.5 Vieta gives $r_1 + r_2 = -\frac{b}{a}$, so with the printed r_1 ,

$$r_2 = -\frac{b}{a} - \frac{a^2 - ab - b}{2a} = \boxed{-\frac{a^2 - ab + b}{2a}}$$

and eliminating r_2 from $r_2 - r_1 = b - a$ gives $2r_1 = -\frac{b}{a} - (b - a) = \frac{a^2 - ab - b}{a}$, which is the printed result.

6.6 For a, r_1, b to be arithmetic, $r_1 - a = b - r_1$, and substituting the printed r_1 gives

$$\frac{b(2a+1)}{a} = 0 \implies b = 0 \text{ or } a = \boxed{-\frac{1}{2}}$$

6.7 With $b = 0$: the sequence $a, r_1, 0, r_2, c$ is symmetric about zero, so $r_1 = \frac{a}{2}$, $r_2 = -\frac{a}{2}$

and $c = -a$. Vieta's product then gives $r_1 r_2 = \frac{c}{a}$, that is $-\frac{a^2}{4} = -1$, so $a = \pm 2$:

$$\boxed{Q(x) = 2x^2 - 2},$$

$$\boxed{Q(x) = -2x^2 + 2}$$

with sequences $2, 1, 0, -1, -2$ and $-2, -1, 0, 1, 2$.

6.8 With $a = -\frac{1}{2}$, r_1 is the mean of a and b :

$$r_1 = \frac{-\frac{1}{2} + b}{2} = \boxed{\frac{2b - 1}{4}}$$

6.9 Substituting r_1 into $Q(r_1) = 0$ with $a = -\frac{1}{2}$ gives $c = \frac{4b + 1}{2}$ and then $4b^2 + 20b + 5 = 0$:

$$b = \boxed{\frac{-5 \pm 2\sqrt{5}}{2}},$$

$$c = \boxed{\frac{-9 \pm 4\sqrt{5}}{2}}$$

Decimals earn nothing here — the question says *exact values*.

§ 7. The greatest sum

7.1 The terms are positive to the 24th and zero at the 25th, so $S_{24} = S_{25}$ and both are greatest:

— —

$$S_{25} = \frac{25}{2}(60 + 0) = \boxed{750}$$

7.2 $u_1 = 80$ and $u_{11} = 16.8$, so $10d = -63.2$ and $d = -6.32$. The terms stay positive while $80 - 6.32(n - 1) > 0$, that is $n < 13.66$, so $n = 13$:

$$S_{13} = \frac{13}{2}(160 - 6.32 \times 12) = \boxed{547.04}$$

A table of S_n finds it faster, but the value must be taken by substituting $n = 13$: the markscheme rejects the graph's vertex $(13.1582\dots, 547.119\dots)$ by name.

§ 8. n has to be a whole number

8.1 Two tables side by side:

n	$P_3(n)$		m	$P_5(m)$
19	190		11	176
20	210		12	210
21	231		13	247

$\boxed{210}$

the $\boxed{20}$ th triangular number and the $\boxed{12}$ th pentagonal number. The markscheme

gives all five marks for 210 seen anywhere.

8.2

$t_n = 2, 7, 15, 26, 40, 57, 77, 100, 126, 155, 187, 222, \boxed{260}$, and $260 = 5 \times 52$, so $\boxed{13}$ rows and $\boxed{5}$ packs. Solving $t_n = 52$ instead is marked M0A0; answering 5 instead of 13 loses the mark too.

8.3 $c = -\frac{m^2}{m+2}$ is an integer when $m+2$ divides m^2 . Since $m^2 = (m-2)(m+2) + 4$, that means $m+2$ divides 4, so $m \in \{-1, -3, 0, -4, 2, -6\}$. Discarding $m=0$ and checking $r = -c/m$ integer as well leaves three, one of which is the given $L(x) = -x - 1$:

$$\boxed{L(x) = -4x + 8},$$

$$\boxed{L(x) = -3x + 9}$$

with sequences $-4, 2, 8$ and $-3, 3, 9$.

§ 9. Terms that are logarithms

9.1(i) $p \ln x - \ln x = \frac{1}{3} \ln x - p \ln x$. Since $x > 1$, $\ln x$ divides out: $p - 1 = \frac{1}{3} - p$, so

$$\boxed{p = \frac{2}{3}}.$$

$$\textbf{(ii)} \quad d = \left(\frac{2}{3} - 1\right) \ln x = \boxed{-\frac{1}{3} \ln x}.$$

(iii) $\ln\left(\frac{1}{x^3}\right) = -3 \ln x$ — the minus sign is the mark. Then

$$\begin{aligned} \frac{n}{2} \left(2 - \frac{n-1}{3}\right) \ln x &= -3 \ln x \\ &\implies n(7-n) \\ &= -18 \\ &\implies n^2 - 7n - 18 \\ &= 0 \end{aligned}$$

$$(n-9)(n+2) = 0 \implies \boxed{n=9}$$

The markscheme's second method needs no algebra: the terms are $\ln x, \frac{2}{3} \ln x, \frac{1}{3} \ln x, 0, -\frac{1}{3} \ln x, -\frac{2}{3} \ln x, -\ln x, \dots$, the first seven cancel to nothing, and the 8th and 9th add to $-3 \ln x$.

9.2(i) $\ln 9 = 2 \ln 3$ and $\ln 1 = 0$, so the terms are $9 + 2 \ln 3, 5 + \ln 3, 1$:

$$d = (5-9) + (1-2) \ln 3 = \boxed{-4 - \ln 3}$$

(ii)

$$\begin{aligned}
S_{10} &= 5(2(9 + 2 \ln 3) + 9(-4 - \ln 3)) \\
&= 5(-18 - 5 \ln 3) \\
&= \boxed{-90 - 25 \ln 3}
\end{aligned}$$

Both halves are needed. Dropping the logarithms because $\ln 1 = 0$ loses the $-25 \ln 3$.